Simulating multi-soliton solutions of the nonlinear Schrödinger equation in Bose-Einstein condensates

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Abstract

Bright solitons in a Bose-Einstein condensate are studied using the focusing nonlinear Schrödinger equation. A semi-implicit finite difference method alongside the predictor-corrector approach is validated using stationary and breathing soliton solutions achieving second order convergence with a particle number deviation of 3.64×10^{-9}. Collisions between two bright solitons were then studied with global phase differences imposed of $\Delta\phi \in \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi\}$ which reveals that peak density scales as $2A^2(1 + \cos(\Delta\phi))$ demonstrating continuous transitions between constructive ($\Delta\phi = 0$) and destructive ($\Delta\phi = \pi$) interference. The intermediate phases display suppressed constructive and destructive results which are often overlooked in the literature. These results confirm the particle-like interactions of solitons by illustrating the elastic nature of their collisions, continuously tunable phase differences and soliton collision dynamics have significant uses in quantum computation and optical communications where collision outcomes influence logic gate operations and transmission quality.

I. INTRODUCTION

Nonlinear wave phenomena arise in a variety of fields, ranging from solid-state physics to plasma physics (1). Nonlinear physics captures field behaviour that is impossible in linear systems such as its localized wave solutions, known as solitons. First observed in 1834 by British engineer Scott Russell (2). Unlike standard results from wave interactions, solitons do not spread due to dispersion; rather, they propagate as a stabilized local wave packet due to the balance between nonlinearity and dispersion which prevent the wave packet from collapsing or dispersing. Consequently, this balance also causes particle-like interactions as collisions do not have a permanent effect on the wave shape of the soliton (2). A balance between the two creates a shape-preserving wave packet. Soliton solutions can be seen to be bright (supports attractive interaction) or dark (supports repulsive interactions), which correspond to a localized peak or dip on the background density, respectively (3). In the context of this paper, the focus will only be on bright soliton solutions that arise from the attractive interactions of the Gross-Pitaevskii equation.

I.A. Background

One of the central models behind nonlinear wave physics is the nonlinear Schrödinger equation (NLSE). Early studies of the nonlinear Schrödinger equation include Vladimir Zakharov's (4) analysis of deep water waves. This same equation was then used, due to a strong connection, by the physicists Gross and Pitaevskii to develop its use in low temperature bosonic gases (5; 6) which allowed them to independently develop the Gross–Pitaevskii equation (GPE) from the nonlinear Schrödinger equation. The use of bosonic gas allows for the use of a macroscopic wavefunction due to Bose-Einstein statistics, stating that bosons can exist in the same quantum state. The GPE uses the macroscopic wavefunction to understand how the wavefunction evolves over time. The use of mean field theory (GPE) then allows for an analysis of the dynamics in a BEC. The time-dependent 1D Gross-Pitaevskii equation is given by:

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + \frac{2\hbar^2 a_s}{ml_\perp^2}|\psi|^2\psi \tag{1}$$

Where ψ is the macroscopic wavefunction, m is the mass, a_s is the s-wave scattering length (interactions between atoms) and $l_\perp = \sqrt{\frac{\hbar}{m\omega_\perp}}$ is the transverse harmonic oscillator length with radial trapping frequency $\omega_\perp$ and it is said that $g = \frac{2\hbar^2 a_s}{ml_\perp^2}$. The second term in the RHS equation (1) $|\psi|^2$ arises due to interatomic interactions characterized by the s-wave scattering length and is known as the nonlinear term. This introduces the nonlinearity, which is responsible for the formation of solitons, as it balances with the dispersion term, forming localized wave packets. The sign of g gives the soliton solutions, determining if the interaction is attractive or repulsive. If $g > 0$, the soliton solutions are said to be dark, or repulsive. If $g < 0$, then the solutions are bright solitons, meaning attractive. In this paper, $g < 0$ is used with known bright soliton solutions moving at a speed of u:

$$\psi(x,t) = \sqrt{\frac{N}{2\zeta_s}}\mathrm{sech}\left(\frac{x-ut}{\zeta_s}\right)e^{if(x,t)+i\phi} \tag{2}$$

Where N is the number of atoms, $\zeta_s = 2\hbar^2/m|g|N$ giving the width of the soliton, ϕ is the global phase, and $f(x,t)$ is a space and time dependent factor (3). The first term of the right-hand side (RHS) of equation (1) represents the kinetic energy of the condensate, which also corresponds to the dispersion. By choosing specific units to remove dimensionality, the Gross-Pitaevskii equation is drastically simplified, along with soliton solutions, which correspond to the standard form of the focusing nonlinear Schrödinger equation. The focusing nonlinear Schrödinger takes the form:

$$i\psi_t = -\frac{1}{2}\psi_{xx} - |\psi|^2\psi. \tag{3}$$

The full derivation of this equation can be found in Appendix A. Solutions of the forms $\psi(x) = \mathrm{sech}(x,t)$ and $\psi\left(x, \frac{n\pi}{2}\right) = 2\mathrm{sech}(x)$ will be tested to determine the accuracy of the numerical code; these correspond to stationary and breathing soliton solutions, respectively. This non-dimensionalisation allows analysis of bright soliton solutions without the complexities introduced by physical units. The 1D Gross-Pitaevskii equation is specifically used because it supports stable bright soliton solutions; in 3D, bright solitons collapse due to the increased role of transverse excitations. Low-energy particles occupy the ground state of the transverse harmonic oscillator, suppressing transverse dynamics and naturally reducing the 3D Gross-Pitaevskii equation to an effective 1D model. The 1D reduction is valid when the energy is low enough for the atom to be confined to the ground state of the transverse harmonic oscillator which reduces transverse oscillations significantly. In 3D the bright solitons collapse so in this instance the quasi-1D approximation is appropriate (3).

I.B. Applications and aims

Having a physical understanding of soliton dynamics allows for further advancements in fields such as optical communications (7). The distance between solitons in multi terabit ultra-high speed optical transmissions within optical fibres is extremely small, understanding the effects of soliton interactions within these systems allows for appropriate adjustments to be made within the optical fibres to account for potential collisions, improving the transmission of the signal (8). Solitons uses also extend to optical computing as it allows for logic operations based on the outcome of soliton-soliton collisions (9). Soliton interactions in computing are modelled using a Manakov system which is a system of two coupled 3-NLS equations describing solitons interactions and behaviour, a further extension of the work presented by this paper. If validation for the numerical framework can be obtained along with theoretical results for the solutions of the 1D Gross-Pitaevskii equation by comparing with experimental observations, it allows for a deeper understanding of BEC dynamics and soliton particle-like behaviour.

Despite the relevance of soliton dynamics, research on stationary bright soliton behaviour and interactions often focus on imposing a phase of $\Delta\phi = 0$ and $\Delta\phi = \pi$ without considering intermediate phase interactions to analyse destructive and constructive interference between solitons (10; 11). Therefore, the aim of this project is to understand how imposing different phases influence soliton collisions for the 1D Gross-Pitaevskii equation in a Bose-Einstein condensate, whilst also testing the validity of the SIFD method to solve the NLSE. This is achieved by validating a semi-implicit finite difference method alongside a predictor-corrector approach with known stationary soliton solutions and then applying it to interacting soliton systems with imposed phase differences to analyse the changes that occur. By combining the numerical validation and the exploration of phase-dependant interactions, a clearer understanding of soliton dynamics in a Bose-Einstein condensate described by the Gross-Pitaevskii equation can be found.

II. METHOD

II.A. Semi-implicit finite difference method and predictor corrector scheme

II.A.i. Semi-implicit finite difference method

The spatial domain is discretized and defined to be $x \in [-L, L]$ to give an evenly spaced grid of N points with equidistant separation defined to be $\Delta x = 2L/(N-1)$, which allows the wavefunction to be written as a discrete vector:

$$\psi(t) = \begin{pmatrix} \psi_1(t) \\ \psi_2(t) \\ \vdots \\ \psi_{N-1}(t) \end{pmatrix} \tag{4}$$

Where $\psi_i(t) \equiv \psi(x_i, t)$. The value of L must be chosen such that boundary conditions $\psi(\pm L, t) \approx 0$ are met within a tolerance. For the code we use a non dimensional value of $L = 10$ which gives a tolerance at the boundary of

$$|\psi(10)|^2 = \mathrm{sech}^2(10) \approx 9.5 \times 10^{-5} \tag{5}$$

Which is negligibly small, confirming this is an appropriate value to use. Observing equation (3), it is evident that a second-order Taylor series expansion can be used to estimate the second spatial derivative of the wavefunction at x_i

$$\left.\frac{\partial^2 \psi}{\partial x^2}\right|_{x=x_0} \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{(\Delta x)^2} + \mathcal{O}(\Delta x^2) \tag{6}$$

With a truncation error of $\mathcal{O}(\Delta x^2)$. This can be represented by an $N \times N$ tridiagonal matrix of the form

$$\left.\frac{\partial^2 \psi}{\partial x^2}\right|_{x=x_0} \approx \frac{1}{(\Delta x)^2} \begin{pmatrix} -2 & 1 & 0 & \dots & 0 \\ 1 & -2 & 1 & \dots & 0 \\ 0 & 1 & -2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -2 \end{pmatrix} \begin{pmatrix} \psi_1(t) \\ \psi_2(t) \\ \vdots \\ \psi_{N-2}(t) \\ \psi_{N-1}(t) \end{pmatrix} \tag{7}$$

Which converts the partial differential equation into a series of 'N' ordinary differential equations across time.

II.A.ii. Predictor-corrector scheme

Now, rearranging equation (3) for the time derivative.

$$\frac{\partial \psi}{\partial t} = -i\left(-\frac{1}{2}\frac{\partial^2 \psi}{\partial x^2} - |\psi|^2\psi\right) \tag{8}$$

The semi-implicit midpoint method is then used to evolve the wavefunction in time by stating that time is advanced from $t^n \to t^{n+1}$ where $t^{n+1} = t^n + \Delta t$ and $n \in \mathbb{Z}$, therefore

$$\psi^{n+1} = \psi^n + \Delta t \frac{\partial \psi^{n+1/2}}{\partial t} \tag{9}$$

Where $\psi^n = \psi(t^n)$. The value $\frac{\partial \psi^{n+1/2}}{\partial t}$ is then approximated using the predictor-corrector approach. This is initialised with the explicit Euler predictor

$$\psi^{(0)}_{n+1/2} = \psi^n + \frac{\Delta t}{2}\mathcal{F}(\psi^n) \tag{10}$$

Where

$$\mathcal{F}(\psi) = -i\left(\frac{1}{2}\frac{\partial^2 \psi}{\partial x^2} - |\psi|^2\psi\right) \tag{11}$$

Which is then corrected at each iteration for $k \geq 1$ by

$$\psi^{(k)}_{n+1/2} = \psi^n + \frac{\Delta t}{2}\frac{\partial}{\partial t}\psi^{(k-1)}_{n+1/2} + \mathcal{O}(\Delta t^2) \tag{12}$$

After approximately 3 iterations, this converges, allowing for an accurate approximation of the wavefunction in time. The convergence was calculated using python to be $\left|\left|\psi^{(k)}_{n+1/2} - \psi^{(k-1)}_{n+1/2}\right|\right| < 10^{-10}$ which is several orders of magnitude less than the truncation error $\mathcal{O}(\Delta x^2)$ indicating that it is appropriate to use, further iterations will have insignificant effects on the final result. The use of a Taylor series for the spatial derivative and a predictor-corrector approach for the time derivative creates a truncation error of $\mathcal{O}(\Delta x^2 + \Delta t^2)$ as second-order approximations are used (17). This method is implemented using a for loop in python along with a sparse linear solver for efficiency to evolve in time. The use of a focusing NLSE is also imperative for solving multi-soliton solutions, as it is integrable. This also causes elastic collisions, giving the solitons their particle-like collision property, which is the primary source for this investigation.

II.B. Stability and choice of method

The semi-implicit finite difference method was tested against a variety of other methods for solving nonlinear equations with $N = 400$ grid points over $T = 1$ with a time step of $\Delta t = 0.001$

to determine its efficiency and error (see Appendix B.)(12; 15).

Method	Error	Time taken (s)
Explicit Euler	N/A	0.907
Implicit Euler	0.2595	5.731
Crank-Nicolson	0.8458	6.830
Semi-implicit finite difference	0.2595	2.919
Split-step Fourier Method	0.2595	0.097

Table I: Comparison of methods to solve wavefunction solutions for the focusing NLSE. Table displays the error in the results and the time taken for the code to run (See Appendix B. for the code)

Table (I) indicates that the use of the semi-implicit finite difference method ensures stable results and a high second order accuracy for both spatial and temporal results (16), which is specifically important for this investigation as it allows for an accurate understanding of soliton behaviour in a BEC. It can be seen that the SIFD method maintains a low error, which is ideal as it will give accurate results. It is not the most efficient method, however. The split step Fourier method (SSFM) is seen to be the most efficient, maintaining the same error as the SIFD, however, performing the calculations approximately 30 times faster. The methods of Crank-Nicolson, explicit and implicit Euler, all perform the same task in significantly more time than SIFD and SSFM, making them far inferior for this use. To determine these results, the NLSE was solved for a stationary soliton solution, the time taken to execute the code was recorded in Python, and analysed. Overall, it is evident that whilst the SIFD method is not the most efficient, it still maintains its accuracy.

III. RESULTS

III.A. Initial results

The stationary, periodic, and moving soliton solutions are used to determine accuracy (Appendix C.) initially, which gives results:

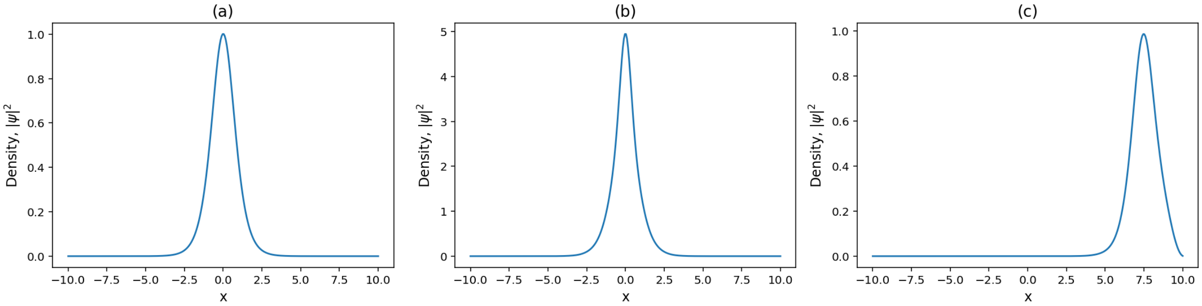

Figure I: Density, $|\psi|^2$, distributions for (a) a single stationary soliton not moving in time, (b) a breathing periodic soliton solution oscillating with a period of $n\pi/2$, and (c) a soliton moving in space with constant velocity.

Figure (I) displays the density distributions solved using the SIFD-PC approach in Python. Figure(Ia) uses soliton solutions $\psi(x, 0) = \text{sech}(x)$, figure (Ib) uses $\psi(x, \frac{n\pi}{2}) = 2\text{sech}(x)$ and figure (Ic) uses $\psi(x, t) = e^{ikx}\text{sech}(x)$ where $v = k = \frac{8\pi}{L}$. The respective space-time diagrams can also be created from the density distributions

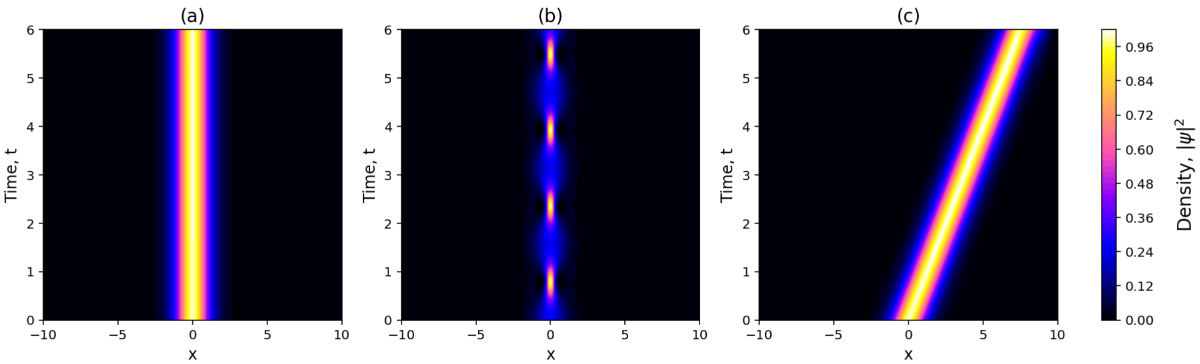

Figure II: Space-time diagrams for (a) a single stationary soliton not moving in time, (b) a breathing periodic soliton solution oscillating with period $n\pi/2$, and (c) a soliton moving in space with constant velocity. Higher density regions indicated by yellow/white and lower density regions indicated by blue/black

The stationary soliton (a) is invariant in time and stays in the same position centred at $x = 0$, the breathing solution (b) oscillates periodically through time, and the moving soliton solutions (c) propagate through time without dispersion. The stability of the solutions are validated by tracking the particle number as follows

$$N = \int_{-\infty}^{\infty} |\psi|^2 dx \tag{13}$$

This was found to have deviation of 3.64×10^{-9} which is extremely small, displaying the stability of the SIFD method. This shows that the second-order approximation used in the method accurately converges at the second order as predicted.

Moreover, breathing soliton solutions were then used to further validate the SIFD approach by analysing the periodicity of oscillations and comparing to expected values

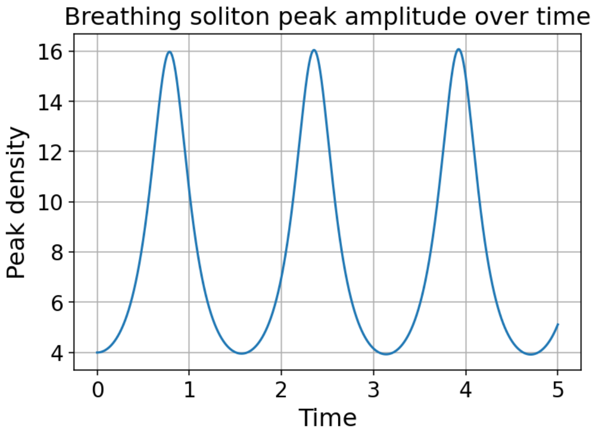

Figure III: Peaks of the periodic soliton solution plotted

The periodicity of the breathing solutions should be of the form $\pi/2$ to match the solution plotted and validate the method. The periods of Fig. (III) were measured to be 1.5678 (4 d.p) and 1.5694 (4 d.p) respectively, which have a respective deviation from $\pi/2$ of 0.001932 and 0.0008710. This small error shows that the SIFD-PC method is accurate for both stationary and breathing soliton solutions. The error in the solutions can be calculated as the root mean squared (RMS), comparing the difference between the numerically calculated solution and the reference solution:

$$RMS = \sqrt{\frac{1}{N_x}\sum_{i=1}^{N_x} |\psi_{num}(x_i) - \psi_{ref}(x_i)|^2} \tag{14}$$

This also allows confirmation of second order spatial convergence by halving the value of the grid spacing, it would be expected that the errors would follow:

$$\mathcal{O}\left(\left(\frac{\Delta x}{2}\right)^2 + \left(\frac{\Delta t}{2}\right)^2\right) = \mathcal{O}\left(\left(\frac{\Delta x^2}{4}\right) + \left(\frac{\Delta t^2}{4}\right)\right) \tag{15}$$

Inferring that the ratio between the original error and scaled error would be a factor of 4. Calculations displayed errors with a ratio of 4.04 for $N_{x1} = 150$ and $N_{x2} = 300$ and an error of 4.01 for $N_{x2} = 300$ and $N_{x3} = 600$ where N_{xn} is the amount of spatial points in the grid. This result confirms second order convergence $\mathcal{O}(\Delta x^2 + \Delta t^2)$.

III.B. Colliding solitons with global phase

Having validated the SIFD approach, by plotting the same density distribution but for a soliton travelling in the opposing direction with a differing symmetric starting point. The initial condition for the soliton collisions follows:

$$\psi(x, 0) = \mathrm{sech}(x - x_0)e^{ikx} + \mathrm{sech}(x + x_0)e^{-ikx}e^{i\Delta\phi} \tag{16}$$

Where k sets the collision velocity and $\Delta\phi$ is the global phase imposed on the second soliton. Using this, the following result is obtained:

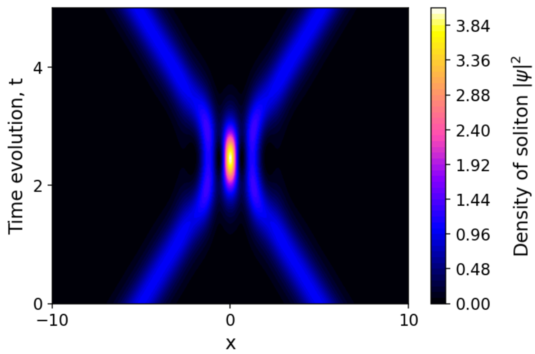

Figure IV: Space-time diagram showing colliding solitons with no phase difference $\Delta\phi = 0$. Both solitons travel with equal and opposite velocities and starting points. Higher density regions indicated by yellow/white and lower density regions indicated by blue/black

The results are then analysed with a global phase of $\pi/2$, π, $3\pi/2$, and 2π imposed on the right-most soliton to see how the interactions change

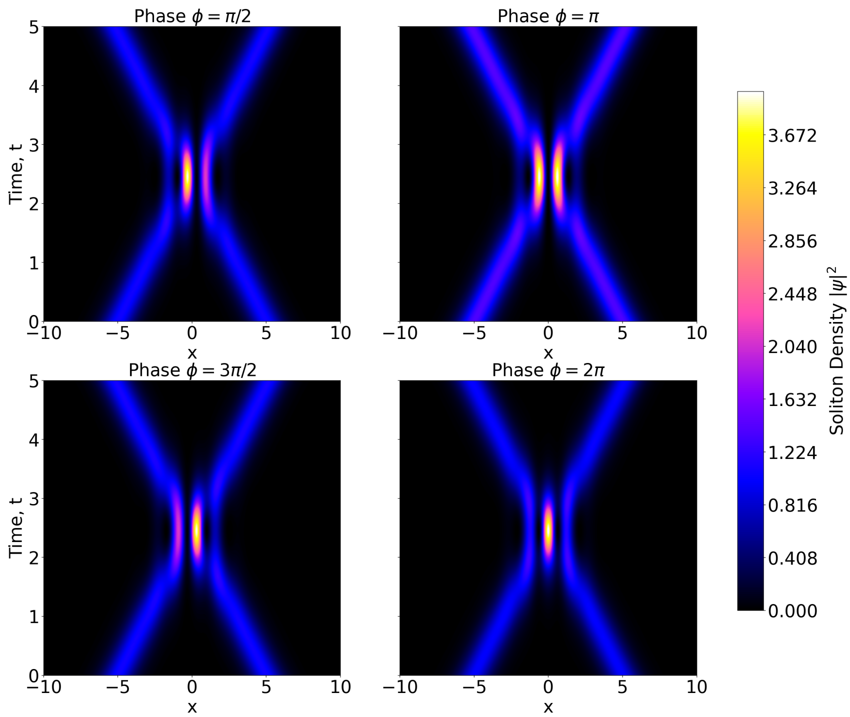

Figure V: Colliding soliton space-time diagrams with global phases imposed of $e^{i\phi}$, $\phi \in \{\frac{\pi}{2}, \pi, \frac{3}{2}\pi, 2\pi\}$. Both solitons travel with equal and opposite velocities and starting points. Higher density regions indicated by yellow/white and lower density regions indicated by blue/black

Figure (V) displays the maximally constructive peaks to be at $\Delta\phi = 2\pi$ and maximally destructive at $\Delta\phi = \pi$. Intermediate peak interferences can be seen in $\Delta\phi = \pi/2$ and $\Delta\phi = 3\pi/2$ which display partially constructive and destructive interference.

IV. DISCUSSION

IV.A. Phase dependant collisions

Figure (IV) displays collision dynamics of two solitons with no imposed global phase $\Delta\phi = 0$ displaying strong constructive interference at the centre as expected. Physically, this displays how densities interact in a nonlinear system (10). The nonlinearity occurs from the nonlinear term, $-|\psi|^2\psi$, as

$$|\psi|^2 = |\psi_1 + \psi_2|^2 = |\psi|_1^2 + |\psi_2|^2 + 2|\psi_1||\psi_2|\cos(\Delta\phi) \tag{17}$$

Where $\Delta\phi = \phi_1 - \phi_2$ and $2|\psi_1||\psi_2|\cos(\Delta\phi)$ is the interference term. This means that as $\phi = 0 = 2\pi$, $\cos(0) = \cos(2\pi) = 1$ creating constructive interference as observed. Conversely as $\phi = \pi$, $\cos(\pi) = -1$ creating destructive interference. Evaluating equation (17) at the point $x = 0$ where both solitons have an equal amplitude $|\psi_1| = |\psi_2| = A$, the peak density simplifies to

$$2A^2(1 + \cos(\Delta\phi)) \tag{18}$$

Equation (18) illustrates that we have a continuous variation in the peak density with a maximum of $4A^2$ at $\Delta\phi = 0$ since $\cos(0) = 1$ and a minimum of zero at $\Delta\phi = \pi$ where $\cos(\pi) = -1$. The intermediate phases $\Delta\phi \in \{\frac{\pi}{2}, \frac{3\pi}{2}\}$ represent suppressed collisions giving values of $2A^2$ since $\cos(\pi/2) = \cos(3\pi/2)$. This confirms that by changing the phase we have a continuously tunable parameter. By finding the peak densities at $x = 0$ from our simulation and plotting against known theoretical densities from equation (18) we can see

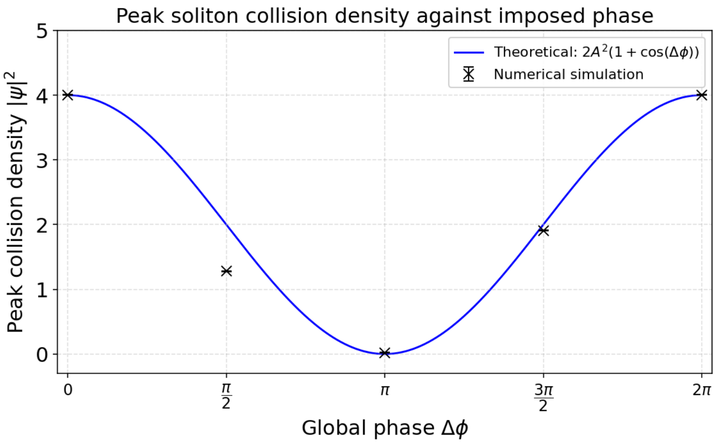

Figure VI: Theoretical values for the density at given phase differences at a position $x = 0$ (blue) modelled using $2A^2(1+\cos(\Delta\phi))$ plotted against obtained peak densities (black) with error bars representing the truncation error on the calculation $\mathcal{O}(\Delta x^2)$.

Figure (VI) display extremely similar values for the peak densities on maximally constructive and destructive phase differences ($\Delta\phi = 0$, $\Delta\phi = 2\pi$ and $\Delta\phi = \pi$). There is a noticeable deviation when a phase of $\Delta\phi = \pi/2$ is reached. This occurs due to a small phase dependant position shift experienced by the solitons caused by the nonlinear term modifying the velocities of the solitons. Equation (18) assumes two equal amplitude solitons and evaluates the density at the maximum when $x = 0$. This change in velocity causes the position shift which makes the densities skew, causing the error which can be seen in Figure(VI). This represents the limitations of the model used.

The literature utilised a degenerate gas of ^{7}Li atoms and imposed a phase on the solitons upon collision. As expected from the obtained results, it can be seen that a phase of $\phi = 0$ creates a strong anti-node, inferring an attractive soliton interaction, and imposing a phase of $\phi = \pi$ displays repulsive soliton interactions, which is in direct accordance with the results found in this paper. Appropriate literature for an experimental analysis can also be used to further validate the results obtained (18), in which it can be seen that there is also a direct correlation between the results obtained in this paper and the literature. Again, the imposed phase results strongly align with the results obtained, inferring that the use of the SIFD and code provides accurate results to analyse soliton behaviour under the focusing NLSE.

While the use of the SIFD approach and 1D GPE to simulate solitary wave collisions has proven to match the phase dependencies found in literature, the approximations used create slight discrepancies between the two. The removal of the transverse mode to reduce the GPE to 1D from 3D drastically simplifies the results; real solitons in BEC's are known to be quasi-1D, meaning that it is approximated to 1D, but there still exist 2 other weakly confined dimensions, which have small effects on the dynamics of the system. Moreover, it is assumed that the solitons exist on a discretized, finite domain $x \in [-L, L]$ with boundary conditions $\psi(\pm L, t) \approx 0$. This is appropriate for an approximation to soliton solutions; however, the solitons never strictly vanish, causing a slight error in the measurement. The truncation error in the SIFD approach with second-order approximations is also a slight limitation to the model; it has been shown that this error is limited due to convergence. These limitations mean that whilst the results provide a good analytical understanding of soliton behaviour, they are not completely analogous to a real model. This can be seen through comparison of the peak strength in the results obtained in this paper and the results obtained by Nguyen, J.H. et al., the simulated overlap regions appear much sharper and more defined than the interactions in experimental observations likely due to simulated results not accounting for atom losses. In real systems imperfections exist causing atom losses which broaden collision dynamics. However, the main physics in phase-imposed soliton collisions is evidently captured and analogous to that of physical systems.

V. CONCLUSION

This project simulated the effects of solitary wave collisions with imposed global phase which was achieved using the semi-implicit finite difference predictor-corrector (SIFD-PC) approach to evolve the known wavefunction in time. The use of the initial condition solutions allowed for the SIFD-PC approach to be validated by testing for conservation of particle number with a deviation of 3.64×10^{-9} along with second order convergence with error ratios of 4.01 and 4.04. To further verify the method analysis of the period of oscillation for breathing solutions was observed. All of the initial condition solutions were correctly validated, showing that the used SIFD approach was appropriate and maintained crucial soliton features. Upon confirmation of the validity of the method, symmetric solitons were studied upon collision. The simulation of colliding solitons revealed the particle-like interactions solitons are known to have, along with displaying the discontinuous jump upon collision.

The primary focus of this paper was to investigate a range of continuous phase differences on soliton interactions $\Delta\phi \in \{0, \pi/2, \pi, 3\pi/2, 2\pi\}$. The collision density was approximated to follow $2A^2(1 + \cos(\Delta\phi))$ with a maxima of $4A^2$ at $\Delta\phi = 0, 2\pi$ and a minima of 0 at $\Delta\phi = \pi$. Intermediate phases $\Delta\phi \in \{\pi/2, 3\pi/2\}$ produced partially attractive and destructive results. Analysis of the obtained peak densities at given phases reveal that the greatest differences from expected results are during intermediate phase differences, which confirm the fact that the nonlinear term modifies the wavefunction velocity. This is a direct characteristic of the nonlinear NLSE.

The results obtained in this paper are in direct accordance with experimental results obtained by Nguyen et al. using ^{7}Li atoms, giving direct confirmation that the SIFD scheme for solving the NLSE maintains the key physics to understand soliton dynamics despite simplifications made, such as assuming a 1D system without transverse oscillations. The findings of this paper allows for use of phase as a tuneable parameter in phase interactions which is directly imperative for its use in logic gates where collision outcomes determine the gate applied in a quantum system. These findings will allow for direct control of logic systems to have precise measurements.

The use of soliton interactions is crucial to understanding nonlinear wave dynamics, such as advancements in matter-wave interferometry or creating logic operations in quantum computing as a result of soliton-soliton collisions. To develop a deeper understanding of solitons, many improvements to the method used today could be made. Whilst the core physics of solitons was maintained, errors from simplifications and assumptions arose in many areas. To create a more accurate simulation to develop a high-resolution understanding of solitons, one could find a way to expand the 1D GPE into 3D to include the transverse mode and fully understand soliton dynamics. Moreover, using a more precise method of calculation could be used to remove the truncation error gained by approximating the solutions to the second order; this would create much more accurate results and help to understand the full scope of nonlinear soliton physics.

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